

Homework 3

Due Friday, February 14

1. Consider the group $SL(2, \mathbf{C})$ of complex 2×2 matrices with unit determinant and the transformation of an Hermitian matrix

$$H = \begin{pmatrix} ct + z & x - iy \\ x + iy & ct - z \end{pmatrix}$$

into another Hermitian matrix H' via

$$H' = M^\dagger H M, \quad (1)$$

where $M \in SL(2, \mathbf{C})$.

- (a) Verify that this transformation preserves the value of $c^2 t^2 - x^2 - y^2 - z^2$.
- (b) Consider the 1-parametric subgroup of $SL(2, \mathbf{C})$ formed by matrices of the form

$$B(b) = \begin{pmatrix} e^b & 0 \\ 0 & e^{-b} \end{pmatrix},$$

where b is a real parameter. Verify that, for such B , $M = B$ in (1) describes a Lorentz transformation (boost) in the z direction and find the velocity of the reference frame that corresponds to such a boost.

- (c) Verify that the multiplication law $B(b)B(b') = B(b + b')$ corresponds to the relativistic law of addition of velocities, for the special case when both velocities are in the z direction.

2. Consider the group $SL(2, \mathbf{R})$, parametrized as follows:

$$M(x, y, \theta) = \begin{pmatrix} \sqrt{1 + x^2 + y^2} - x & y \\ y & \sqrt{1 + x^2 + y^2} + x \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Let us rename the parameters x, y, θ into α_a , $a = 1, 2, 3$, as follows: $x \equiv \alpha_1$, $y \equiv \alpha_2$, $\theta \equiv \alpha_3$.

- (a) Find the generators X_a , $a = 1, 2, 3$, corresponding to this parametrization. Please do not include i in the definition of the generators, so they are real matrices.
- (b) Compute all nonzero structure constants f_{ab}^c of the algebra (defined here by $[X_a, X_b] = \sum_c f_{ab}^c X_c$).

(c) Consider the adjoint representation of the algebra, in which the generators are represented by operators $D(X_a)$ acting as follows:

$$D(X_a)|X_b\rangle = |[X_a, X_b]\rangle.$$

Compute the matrices of $T_a \equiv D(X_a)$ in the basis of $|X_b\rangle$. As a check of your calculation, verify that commutation relations among these matrices correctly represent the algebra.